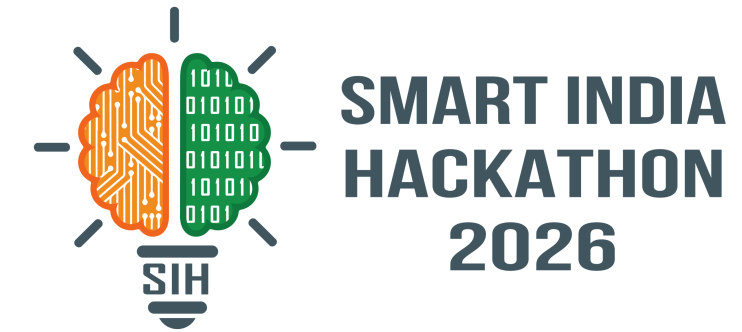


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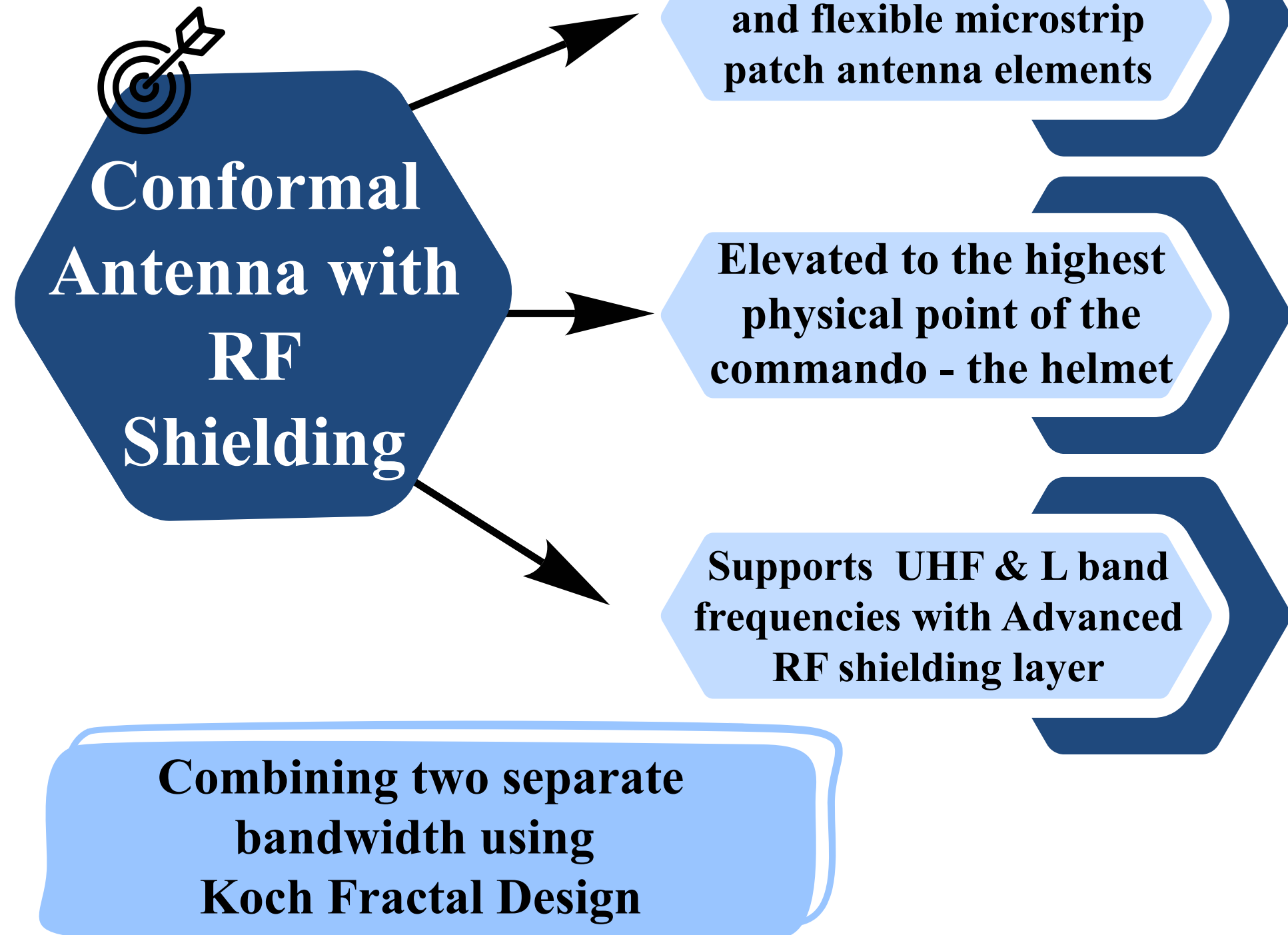
- **Problem Statement ID - SIH26185**
- **Problem Statement Title -** Helmet mounted conformal antenna for tactical communications in urban CQB environments.
- **Theme -** Robotics and drones
- **PS Category -** Hardware
- **Team ID -** 133214
- **Team Name -** HYConA



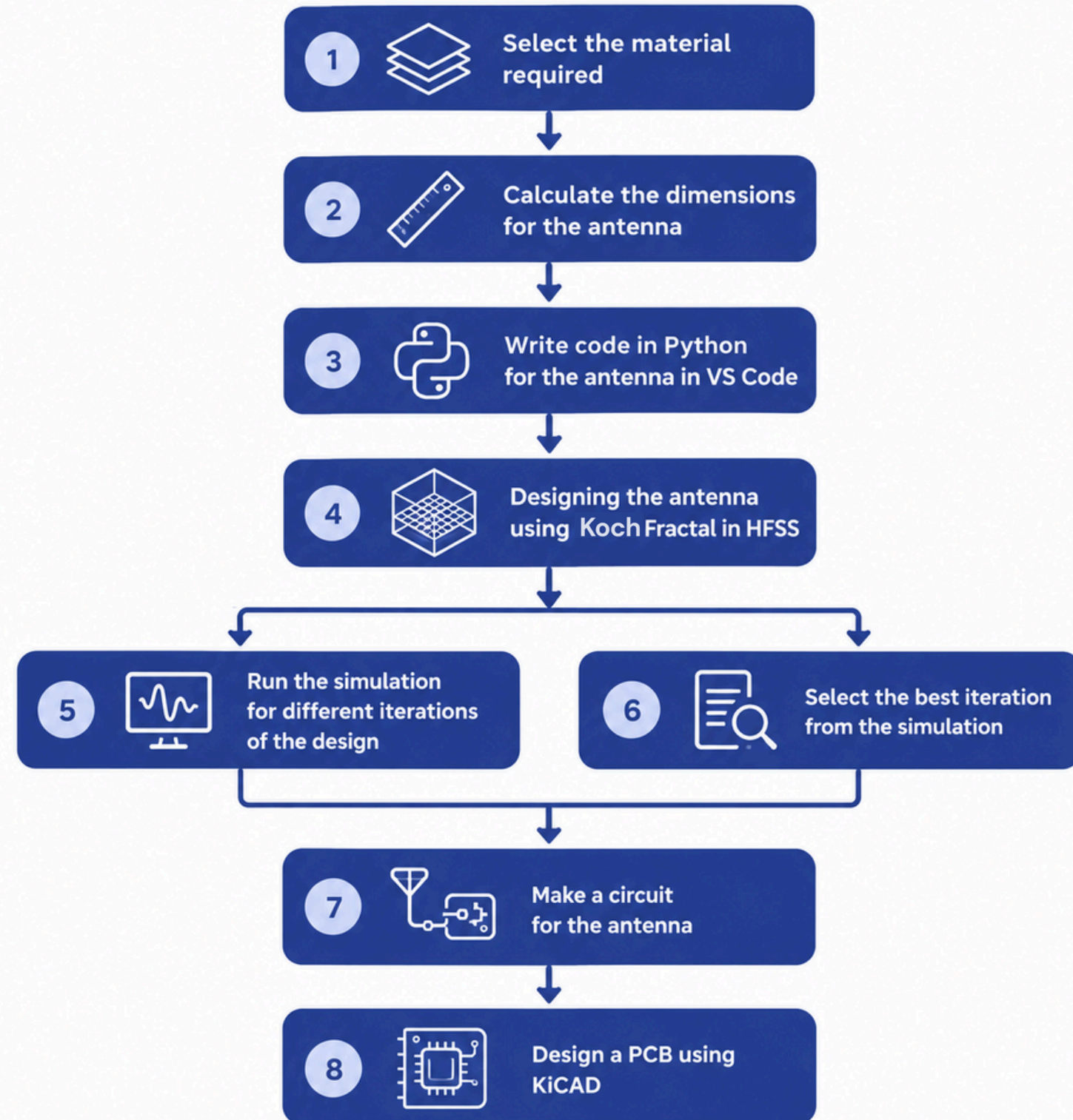
PROBLEM



Solution



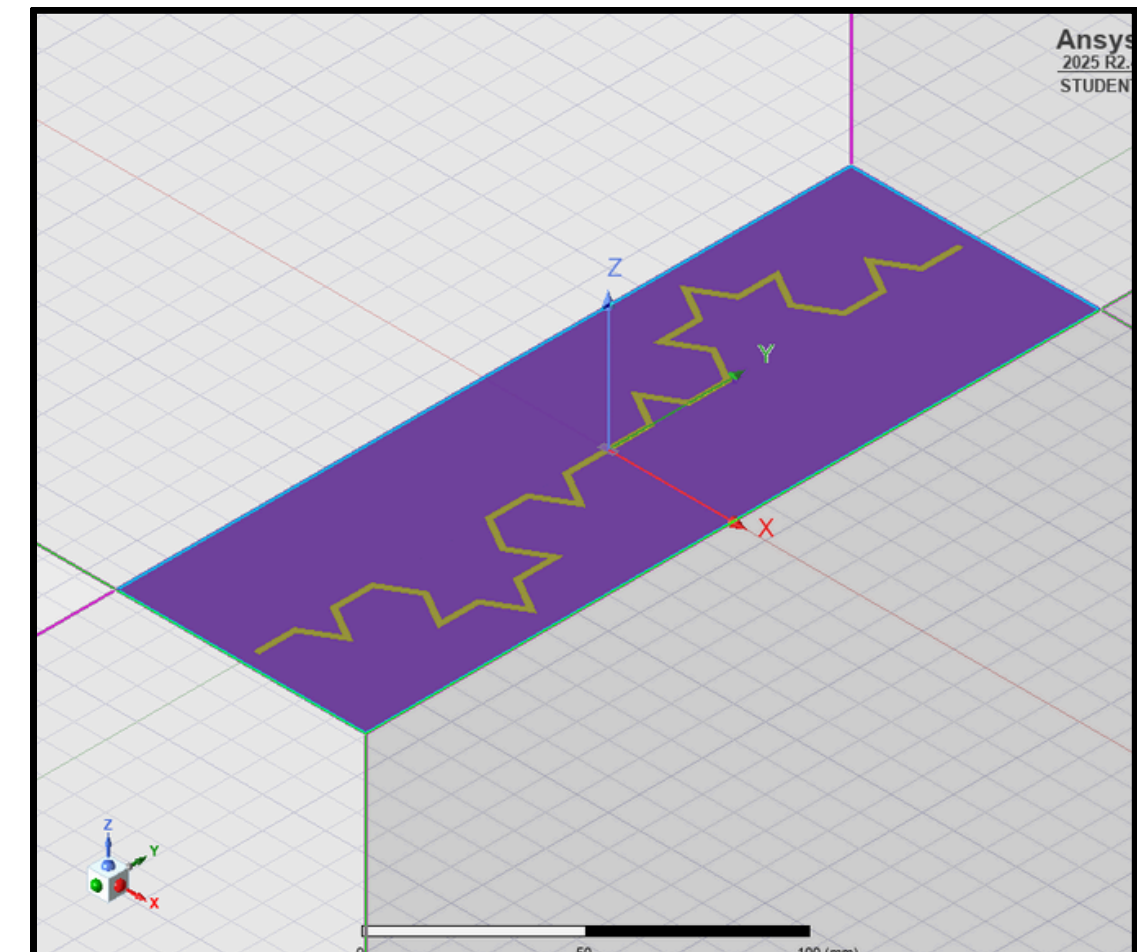
Methodology



Technology Used:

HFSS
VS Code
Python
KiCAD Simulator

Hardware Simulation



Project Links:

[Google Drive](#)
[GitHub Repo](#)



FEASIBILITY

- Mature UHF/L-band patch antenna tech
- Proven in wearable & body-worn comms
- Prototype testable with lab-grade VNA
- Usage of Simple Fabrication Techniques



CHALLENGES

- Gain loss when antenna is bent/curved
- SAR vs radiation-efficiency trade-off
- Must not affect helmet ballistic rating



STRATEGIES

- EM simulation (HFSS) before fabrication
- EBG shielding layer to redirect radiation
- Lightweight ruggedized flexible laminate

Implementation Roadmap - Prototype To Field Ready

Design

Simulate
(EM model)

Prototype
(low-cost)

Bench Test

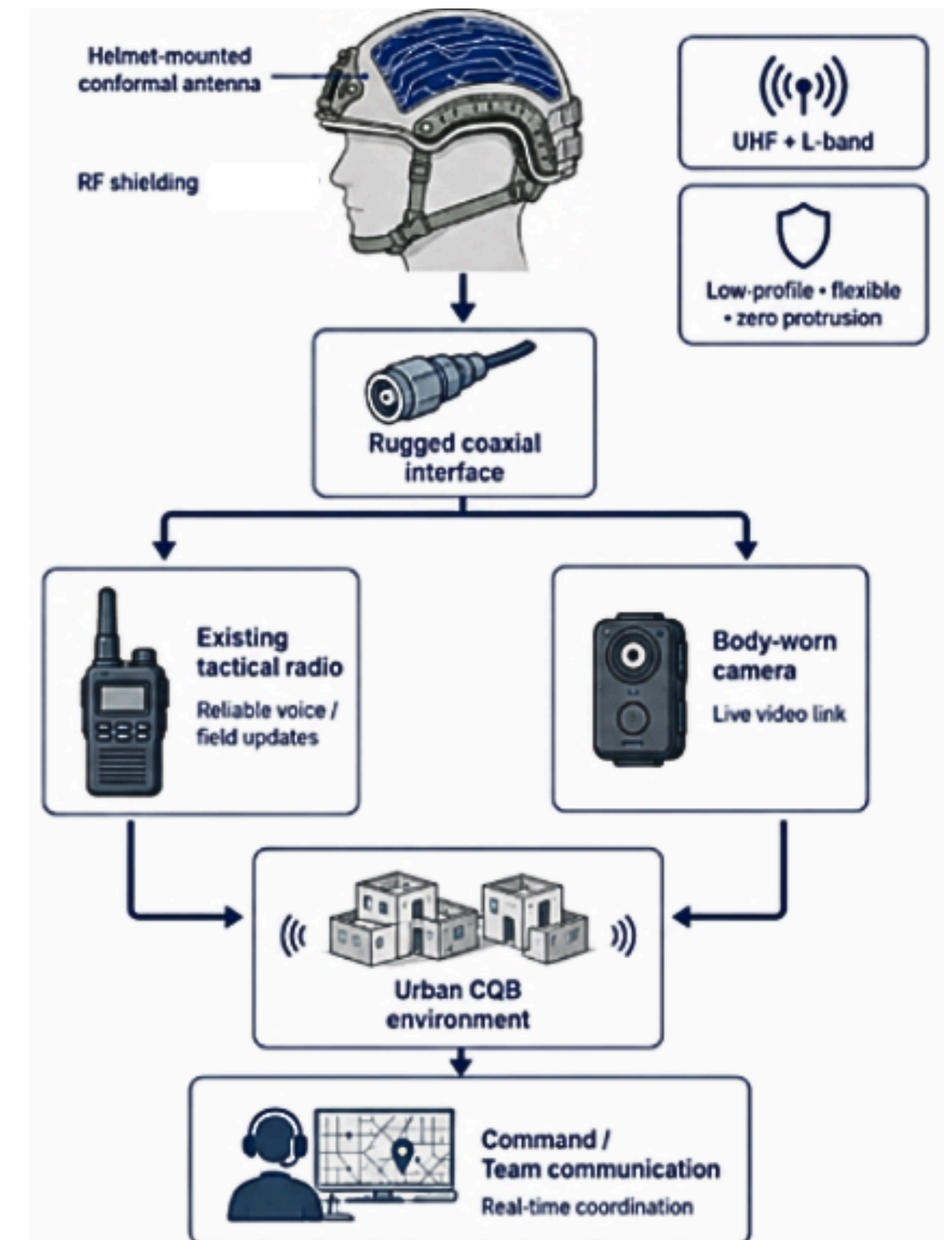
Field
Integrate

TARGET AUDIENCE IMPACT:

- Continuous, reliable radio contact gives uninterrupted real-time field updates
- Optimized Functionality due to no additional hardware on vest

BENEFITS OF THE SOLUTION:

- Tactical communication in which risk of failure is least
- Compatible with in use radios, can use existing infrastructure
- Low cost materials enable cost per unit at scale
- Supports India's defence indigenization, increases security and reduce foreign imports



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- Pooja Naresh Bhatt, Rashmi Pandhare, "[A tensile wearable SHF antenna with efficient communication in defense beacon technology](#)", Defence Technology, Volume 41, 2024, Pages 198-210, ISSN 2214-9147.
- T. Saeidi, I. bin Ismail, V. Saleh and A. R. H. Alhawari, "[Triple band modified 90 degree Koch fractal H-slot microstrip antenna](#)," 2016 6th International Conference on Intelligent and Advanced Systems (ICIAS), Kuala Lumpur, Malaysia, 2016, pp. 1-5, doi: 10.1109/ICIAS.2016.7824131.